Project Synopsis: VendorGuard AI – Evidence-Grounded Supplier Verification & Risk Intelligence

Verify the Supplier. Verify the Evidence. Verify the AI. Major Project Synopsis, Academic Year 2026–2027.

Cover

Project title: VendorGuard AI – Evidence-Grounded Supplier Verification, Predictive Risk Intelligence & AI Reliability Platform

Official problem statement: AI-Powered B2B Supplier Verification and Predictive Trust Scoring Engine

Student / Team: Aditya Yadav – Individual project (team size 1)

Enrollment / Roll number: 2315510016

Institute / Course: GLA University, Mathura – B.Tech (AI & ML), 4th Year (Final Year), Section AC – Major Project

Project mentor: _____

Version: Draft v0.1

Date: 05 Oct 2026

GitHub repository link: https://github.com/adityayadav161003/VendorGuardAI

Revision history

Version	Date	Author	Change
v0.1	05 Oct 2026	Aditya Yadav	Initial scope, methodology, architecture and evaluation plan

1. Overview

Problem statement: AI-Powered B2B Supplier Verification and Predictive Trust Scoring Engine. Businesses onboarding unfamiliar suppliers must review GST, PAN, company-registration, ISO/compliance, bank and invoice documents, cross-check them against external sources and judge risk – a slow, error-prone process that is hard to audit, and generic AI tools can add unsupported claims.

Project summary: VendorGuard AI is an evidence-grounded supplier due-diligence platform intended to help businesses review supplier documents, identify inconsistencies, surface risk indicators, and inspect the evidence behind an assessment. It combines document processing, entity resolution, evidence retrieval, predictive risk modelling, supplier relationship-graph analysis, and claim-level verification of AI-generated explanations. It is a decision-support tool, not an automated legal or procurement authority.

Goal: Design, implement and evaluate a working web platform that produces explainable supplier risk assessments from submitted documents and permitted evidence sources, while identifying unsupported or contradictory claims in its AI-generated reports.

Non-goals:

- No definitive legal finding that a supplier is fraudulent and no guarantee of supplier quality.
- No automatic purchases, payments, contracts or blacklisting.
- No claim of live government-registry verification unless an authorized source/API is actually available.
- No paid proprietary AI API as a mandatory core dependency.

Value proposition: A single workflow for supplier document review, evidence-backed risk indicators, relationship analysis and transparent AI explanations. Material factual claims should link to evidence or be labelled contradicted, unsupported, or unverifiable.

2. Scope and Control

2.1 In-scope

- Supplier profiles and controlled user access.
- PDF/image intake, OCR, document classification and field extraction.
- Cross-document consistency checks and entity resolution for names, addresses, identifiers and dates.
- Evidence store with source references and observation timestamps.
- Claim-level verification: supported, contradicted or insufficient evidence.
- Explainable risk indicators and a documented baseline model where suitable labels exist.
- Supplier relationship graph and suspicious-pattern analysis; GNN as an optional experiment.
- Web dashboard, evidence-linked reports, supplier comparison, REST API, testing and documentation.

2.2 Out-of-scope

- Automated final credit/legal/procurement decisions.
- Guaranteed access to every registry or confidential transaction database.
- Production ERP integrations, paid data brokers and training an LLM from scratch.
- Real-world fraud-accuracy claims without representative labelled data.

2.3 Assumptions

- Users are authorized to submit the documents they upload.
- External sources are used only where access and terms permit.
- Initial experiments may require public, curated and clearly labelled synthetic data.
- Risk outputs are decision-support indicators requiring human review.

2.4 Constraints

- Solo development alongside academic work.
- Limited compute and potentially limited labelled supplier-fraud data.
- OCR quality varies by scan quality, layout and language.
- Public sources can be incomplete, stale or restricted.

2.5 Dependencies

- Python and open-source AI/ML libraries.
- Database and protected document storage.

- Permitted source material and curated evaluation fixtures.
- Optional local/open LLM; rules and conventional ML baselines remain available.

2.6 Acceptance criteria and sign-off

ID	Acceptance criterion	Verification
AC1	Supported PDF/image uploads are stored and processed or fail safely.	Upload tests
AC2	Extracted fields show source page/snippet and confidence where available.	Labelled field test set
AC3	Known cross-document conflicts are flagged with both sources.	Conflict fixtures
AC4	AI claims are labelled supported, contradicted or insufficient evidence with evidence links.	Claim benchmark
AC5	Reports show risk indicators, explanations, evidence and limitations.	UI/report review
AC6	Evaluation includes baselines, error analysis and reproducible metrics.	Evaluation report

Sign-off: Mentor approves scope, demo and evaluation report; core acceptance criteria AC1–AC6 pass.

Sign-off table

Stakeholder	Role	Decision area	Signature/Approval	Date
_____	Project mentor	Scope, final acceptance	_____	_____
Aditya Yadav	Student / project lead	Release readiness	_____	_____

3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements/scope	Student	Student	Mentor, potential users	Department
Architecture/data	Student	Student	Mentor	Department
Implementation	Student	Student	Mentor/peers	Department
Testing/evaluation	Student	Student	Mentor	Department
Security/privacy	Student	Student	Mentor	Department
Final demo/docs	Student	Student	Mentor	Department

4. Team and Roles

Member	Role	Responsibilities	Key skills	Availability	Contact
Aditya Yadav	Individual project lead	Research, architecture, coding, data prep, modelling, testing, documentation and demo	Python, ML, NLP, document AI, graph analysis, full-stack development	3 months	_____

5. Week-wise Plan and Assignments

(Indicative 16-week plan; replace with official semester dates and mentor milestones. All work is owned by the student as an individual project. Figure 6 shows an extended 6–8 month roadmap for the full platform vision; this synopsis commits to the 16-week core scope, with GNN experiments, monitoring and integrations deferred if time is short.)

Week	Milestone	Main work (Aditya)	Deliverables	Status
1	Requirements	Mentor scope review; identify use cases and non-goals; user discovery if feasible.	Approved scope	Planned
2	Literature/data plan	Review supplier risk, document AI, claim verification and graph fraud; assess dataset access.	Literature matrix and data plan	Planned
3	Architecture/setup	Repository, API skeleton, schema, configuration, tests and logs.	Running skeleton and diagrams	Planned
4	Document intake	File validation, metadata, PDF/image parsing and storage.	Upload workflow	Planned
5	OCR/extraction	OCR and extraction of selected fields; preserve source references.	Extraction baseline	Planned
6	Entity resolution	Normalize fields and detect cross-document conflicts/expiry.	Consistency module	Planned
7	Evidence retrieval	Index passages and store source metadata/timestamps.	Evidence API	Planned
8	Claim verification	Claim extraction and support/contradiction/insufficient-evidence classification.	Claim benchmark baseline	Planned
9	Risk baseline	Define interpretable features; train only if labels are defensible, otherwise use rules.	Baseline report	Planned
10	Graph intelligence	Build relationship graph and test shared-entity/anomaly signals.	Graph view and synthetic tests	Planned
11	Explainability	Combine indicators and evidence into reviewable reports.	Traceable report	Planned
12	Frontend/API integration	Dashboard, supplier profile, document review and comparison.	End-to-end workflow	Planned
13	Security/robustness	Access control, upload checks, secrets, audit trail and error handling.	Security checklist	Planned
14	Evaluation	Held-out tests, baselines, latency, false-positive/negative analysis.	Results and limitations	Planned
15	Deployment/user review	Deploy demo if feasible; test usability and fix priority issues.	Demo and test report	Planned
16	Finalization	README, diagrams, reproducibility, report, slides and demo script.	Final submission package	Planned

6. Users and UX

6.1 Personas

SME procurement owner: Reviewing an unfamiliar supplier before a purchase.

Procurement/operations staff member: Checking missing, expired or conflicting documents.

Manager/reviewer: Needs the evidence and uncertainty behind risk indicators.

6.2 Top user journeys

Onboarding review: Create supplier profile → upload documents → inspect extracted fields → review conflicts → generate evidence-backed report.

Report review: Open report → inspect risk indicators → follow evidence links → record human review decision.

AI analyst query: Ask AI analyst a question → retrieve evidence → verify material claims → return a cited answer with uncertainty labels.

6.3 User stories

- As a procurement user, I want extracted fields linked to sources so I can verify them.
- As a reviewer, I want document conflicts and expiry dates highlighted.
- As a user, I want factual AI statements backed by evidence.
- As a manager, I want comparable risk indicators without treating scores as definitive verdicts.

6.4 Accessibility & localization

The initial interface targets English, keyboard-operable core workflows, readable contrast, clear status labels and a responsive desktop-first layout.

7. Market and Competitors

7.1 Competitor table

Category	Typical capabilities	Potential gap / hypothesis	VendorGuard differentiation
Enterprise supplier-risk platforms	Monitoring, compliance workflows, risk dashboards, integrations	May be complex or costly for some SMEs; validate through interviews	Focused supplier-onboarding workflow
OCR/document extraction tools	Extract fields from PDFs/images	Extraction alone does not establish identity or reconcile records	Cross-document checks plus provenance
Generic LLM/RAG apps	Answer questions over documents	Can generate unsupported claims	Independent claim verification and uncertainty labels
Fraud/graph analytics	Shared-entity and network signals	May require proprietary transaction data	Transparent graph signals on permitted/synthetic data

7.2 Positioning

Unique angle: An affordable, evidence-first supplier due-diligence assistant for small and medium-sized businesses.

Validation status: Positioning hypothesis only – demand and willingness to pay are not yet proven.

7.3 Validation plan

- Interview 3–5 prospective users initially and expand if feasible.
- Ask about onboarding steps, document types, review time, common failure cases and decision owners.
- Test whether evidence-linked reports solve a recurring problem and earn user trust.
- Use synthetic/anonymized data in demos unless permission for real records is obtained.

8. Objectives and Success Metrics

Objective	Metric	Evaluation plan
Document extraction	Per-field precision, recall, F1	Label test documents; report field-level errors.
Consistency detection	Precision, recall, F1	Compare rules with normalization/entity-resolution pipeline.
Claim verification	Per-class precision/recall, macro-F1	Evaluate supported, contradicted and insufficient-evidence classes.
Risk model	ROC-AUC/PR-AUC where appropriate; calibration/confusion matrix	Use only when labels and sample size are defensible; compare simple baselines.
Graph signals	Precision@k/recall on labelled or planted cases	Separate synthetic benchmark results from real-world validity.
Explainability	Share of material claims linked to evidence or uncertainty labels	Audit generated reports.
Usability	Task completion and user feedback	Small task-based review; set thresholds after pilot.
Reliability	Latency, failures, reproducibility	Benchmark on documented hardware and sample sizes.

No model-performance results are claimed in this synopsis; all outcomes will be measured after implementation.

9. Key Features

Feature	Description	Priority	Acceptance criteria
Supplier profiles	Business details, documents, status and review notes.	Must	Create/view/edit profile and audit changes.
Document intake	Validate supported PDF/images and file limits.	Must	Valid files accepted; invalid files fail safely.
OCR/extraction	Extract configured fields with page references.	Must	Field metrics and source links displayed.
Consistency checks	Compare fields and detect conflicting/expired values.	Must	Known test conflicts flagged.

Feature	Description	Priority	Acceptance criteria
Evidence retrieval	Search uploaded docs and permitted source snapshots.	Must	Results show source ID and timestamp.
Claim verification	Label claims supported/contradicted/insufficient evidence.	Must	Test claims return label and evidence.
Risk indicators	Transparent rules/features with explanations.	Must	Every indicator has definition and reason.
Graph intelligence	Map entities and shared relationships.	Should	Inspectable graph and flagged patterns.
AI analyst	Answer evidence-grounded questions.	Should	Citations or uncertainty on material claims.
Comparison/export	Compare suppliers and export report.	Could	Report includes evidence and limitations.
Monitoring/API	Refresh permitted sources and expose endpoints.	Could	Only implemented sources are labelled active.

10. Architecture

10.1 High-level

Clients: React + TypeScript dashboard (supplier workflow, report visualization, graph view).

Services: FastAPI backend (authentication, validation, audit) → document intake → OCR/parsing → field extraction → entity resolution → supplier profile + evidence store → evidence retrieval → claim verification | risk model/rules | supplier graph.

Outputs: Explainable risk report, evidence-linked AI analyst and human review.

Data stores: PostgreSQL (profiles, metadata, claims, evidence, assessments), FAISS vector index, protected document storage.

Integrations: Permitted external sources only (e.g., MCA/GST portals, public registries, news, sanctions lists) where access and terms allow.

10.2 Proposed technology components

Layer	Candidate technology	Purpose
Frontend	React, TypeScript, Tailwind CSS	Supplier workflow and report visualization
Backend	Python, FastAPI, Pydantic	Business logic, validation and APIs
Document AI	PyMuPDF/pdfplumber, PaddleOCR or Tesseract	PDF/image text extraction
NLP	spaCy, Transformers, rules	Entity/field/claim extraction
Evidence retrieval	Sentence embeddings, FAISS, PostgreSQL metadata	Retrieve source passages and preserve provenance
Claim verification	NLI model plus rules/evidence thresholds	Support/contradiction/insufficient-evidence labels

Layer	Candidate technology	Purpose
Risk modelling	Scikit-learn; optional XGBoost/LightGBM	Interpretable prediction where labelled data supports it
Graph analysis	NetworkX; optional PyTorch Geometric	Relationships and suspicious-pattern experiments
Database	PostgreSQL	Profiles, metadata, claims, evidence and assessments
Deployment	Docker; local first, hosting if feasible	Reproducible demo

10.3 API spec snapshot

Endpoint	Method	Purpose
/api/suppliers	POST	Create supplier profile
/api/suppliers/{id}/documents	POST	Upload document
/api/suppliers/{id}/analyze	POST	Start analysis job
/api/suppliers/{id}/report	GET	Retrieve evidence-linked report
/api/claims/verify	POST	Verify claims against evidence
/api/suppliers/compare	POST	Compare selected suppliers

10.4 Config and secrets

Keep secrets in environment variables, not Git; commit .env.example only. Validate file types/sizes, store uploads outside public paths and restrict access to uploaded files. Avoid logging raw documents or sensitive identifiers.

11. Data Design

11.1 Data dictionary

Entity	Key fields	Notes
User	user_id, name, email, role, organization, created_at	Store minimum necessary personal data.
Supplier	supplier_id, name, business_type, industry, website, address, status, created_by	Access-control sensitive identifiers.
Document	document_id, supplier_id, document_type, file_path, extracted_text, metadata, uploaded_by, status	Protected file storage and metadata.
ExtractedEntity	entity_id, document_id, entity_type, entity_value, confidence_score, page_number	Keep source provenance.
EvidenceSource	evidence_id, source_type, title, url, snippet, retrieved_at, reliability_score	Record origin and observation time.
AnalysisResult	analysis_id, supplier_id, risk_score, risk_level, summary, model_version, generated_at	Record model/rule version.
Claim	claim_id, analysis_id, document_id, claim_text, claim_type, status, confidence_score	Supported / contradicted / insufficient evidence / review.

Entity	Key fields	Notes
ReviewerAction	review_id, claim_id, reviewer_id, action, comment, reviewed_at	Human decisions stored separately from model output.
RiskReport	report_id, supplier_id, generated_by, overall_risk_score, key_findings, recommendation, report_url	Final evidence-linked supplier report.

11.2 Schemas and migrations

Use versioned migrations and keep raw documents, extracted fields, evidence, claims and assessments logically separate. Store model/rule versions for reproducibility.

11.3 Privacy, retention, backup/DR

Prefer synthetic/anonymized demo data. Obtain permission before processing real supplier records and define deletion procedures. Use role-based access and protect files in transit and at rest where deployed. Document retention/backup settings; do not claim recovery targets until tested.

12. System Design Diagrams

12.1 State Transition Diagram (claim verification)

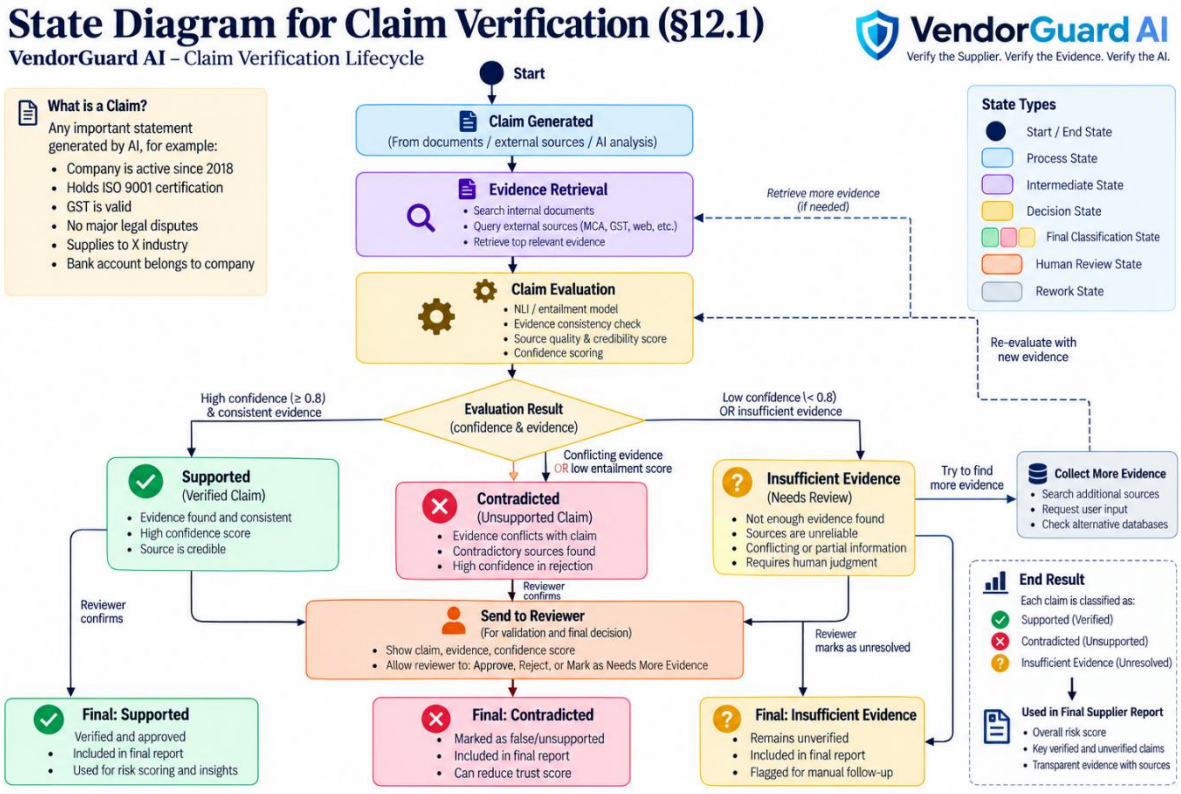


Figure 1: Claim verification state diagram

Claim → retrieve candidate evidence → assess relevance → apply rules/NLI → supported / contradicted / insufficient evidence → attach sources and model version → reviewer accepts, rejects or requests more evidence.

12.2 Sequence Diagram

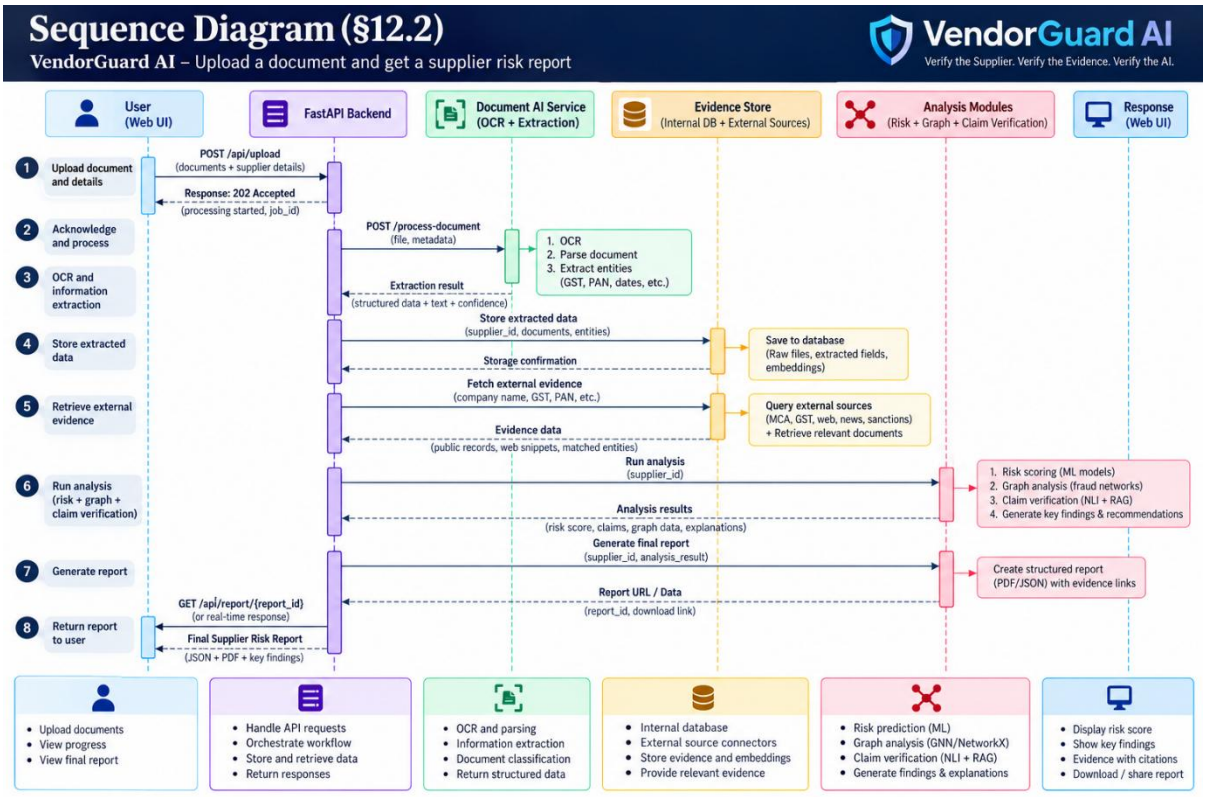


Figure 2: Sequence diagram – upload a document and get a supplier risk report

The user uploads documents and supplier details; the FastAPI backend returns a job ID, calls the document AI service for OCR and extraction, stores the results in the evidence store, retrieves external evidence, runs the analysis modules (risk scoring, graph analysis and claim verification) and returns an evidence-linked supplier risk report to the web UI.

12.3 Use Case Diagram

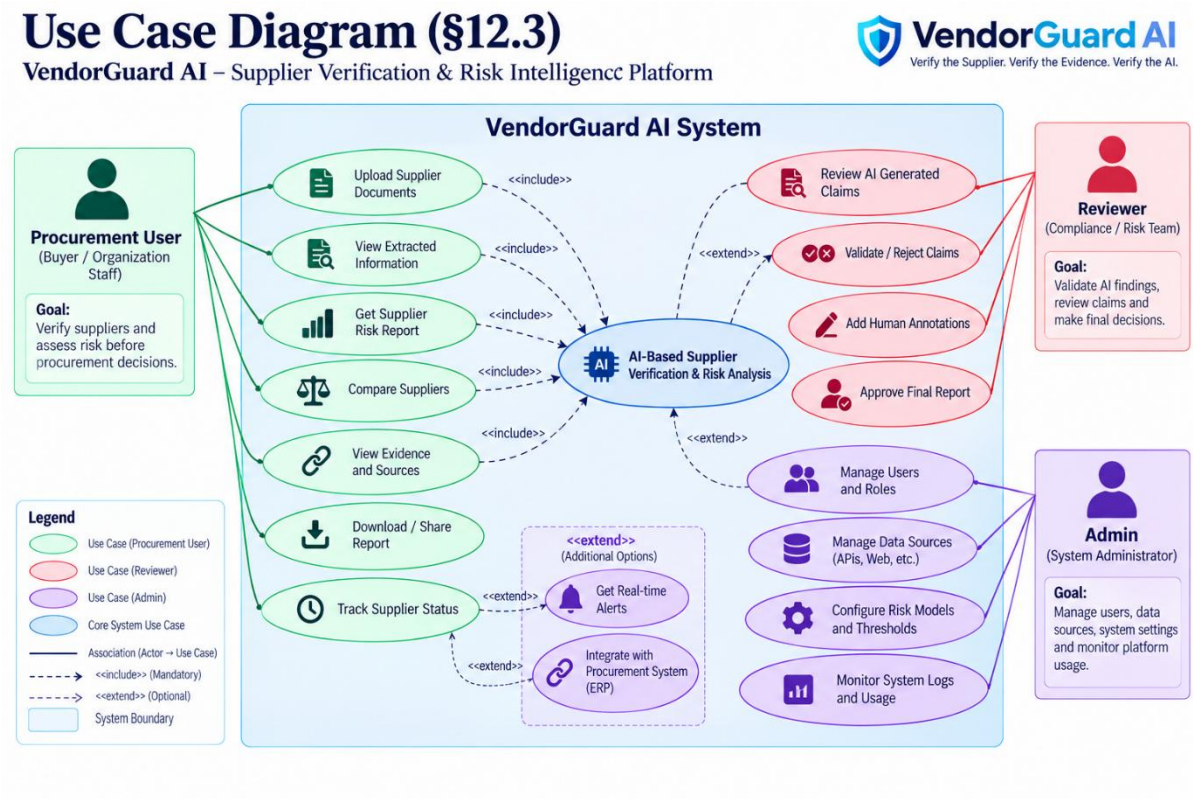


Figure 3: Use case diagram

- Procurement user: manage supplier, upload documents, run analysis, inspect report and compare suppliers.
- Reviewer: inspect evidence, resolve false matches and record review decision.
- Administrator: manage access, audit events and permitted source connectors.

12.4 Data Flow Diagram

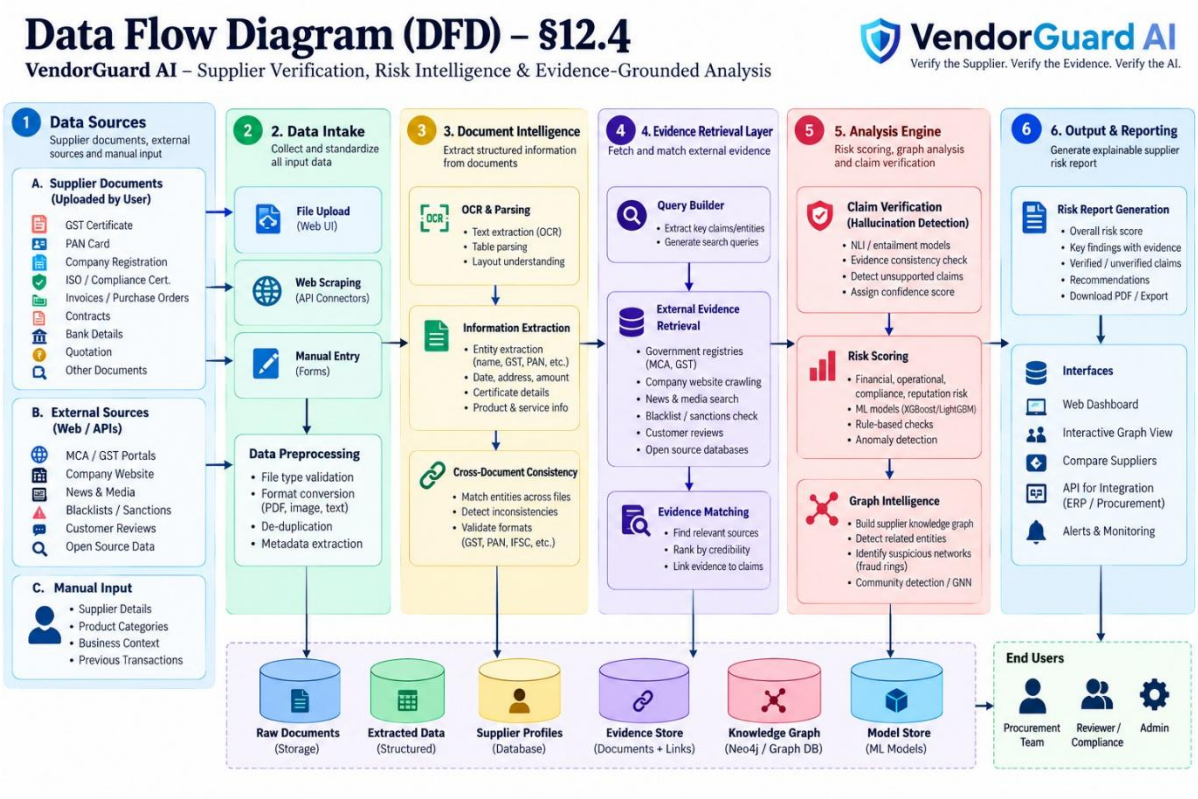


Figure 4: Data flow diagram

External inputs (documents, user-entered details, permitted sources) → intake/validation → extraction/normalization → supplier/evidence records → analysis modules → report/API output. Human corrections are stored separately and are not assumed to be ground truth.

12.5 ER Diagram (logical)

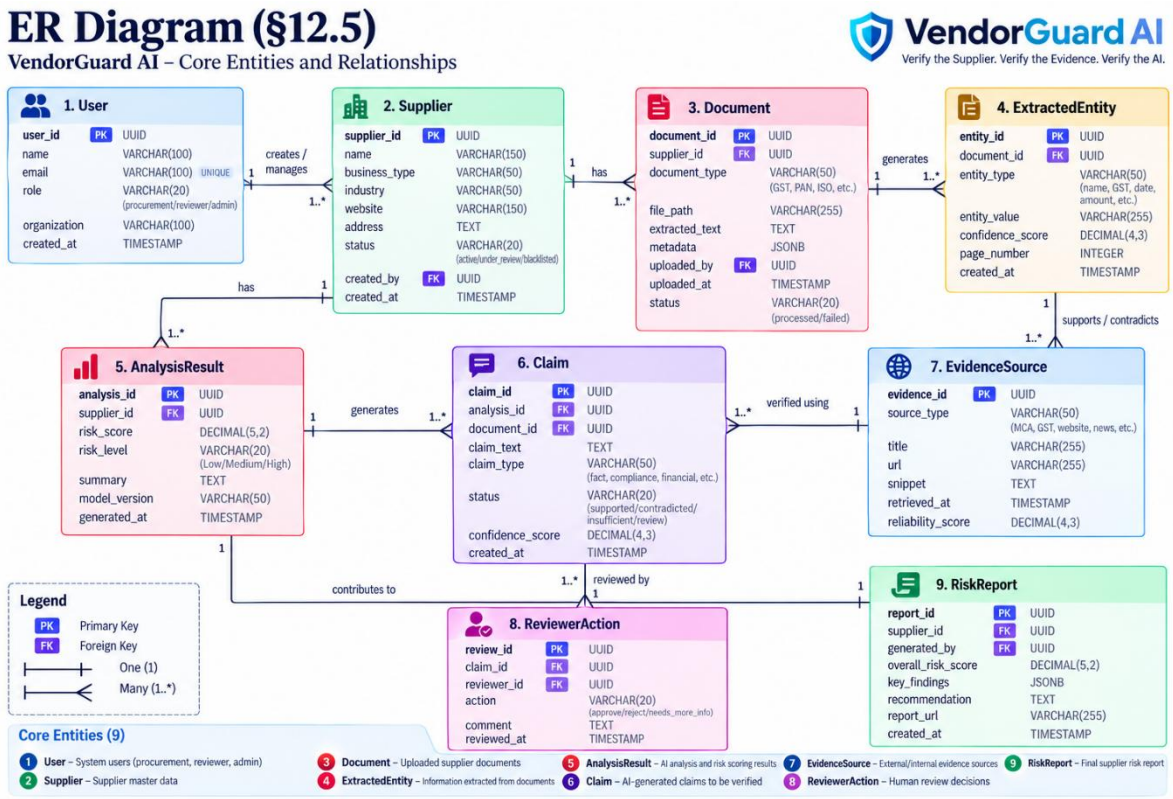


Figure 5: ER diagram – core entities and relationships

User 1:N Supplier (creates/manages); Supplier 1:N Document; Document 1:N ExtractedEntity; ExtractedEntity 1:N EvidenceSource (supports/contradicts); Supplier 1:N AnalysisResult; AnalysisResult 1:N Claim; Claim N:1 EvidenceSource (verified using); Claim 1:N ReviewerAction; AnalysisResult 1:1 RiskReport. Audit events and supplier-graph relationships are stored alongside these core tables.

12.6 Technical Workflow (summary)

User → supplier profile → document upload → validation/OCR → extraction → entity normalization → consistency checks → evidence retrieval → claim verification → risk/graph analysis → report → human review.

12.7 Work Architecture Diagram



Figure 6: VendorGuard AI platform architecture, end-to-end flow and sample output

Start as a modular monolith with separated Python modules and API boundaries to reduce solo-developer overhead. Split services only if scale or deployment needs justify it. The platform overview above shows the six layers (data ingestion, document intelligence, knowledge & evidence, AI analysis engine, reasoning & AI agent, outputs & interfaces), the end-to-end flow and a sample supplier risk output.

13. Quality: NFRs and Testing

13.1 Non-functional requirements

Area	Requirement	Measurement
Reliability	Invalid/unsupported files fail safely.	Negative tests and logs
Performance	Measure processing latency and API p50/p95 on stated hardware.	Repeatable benchmark
Security	No committed secrets; authenticated access; validated uploads.	Scans and access tests
Traceability	Claims link to evidence or uncertainty state.	Automated report audit
Reproducibility	Pinned dependencies, data versions and documented commands.	Clean environment run
Maintainability	Modular code, schemas, README and tests.	Code checklist
Usability	Users can understand flagged conflicts and indicators.	Task-based feedback

13.2 Test plan

Area	Type/tools	Exit criteria
API	pytest, FastAPI TestClient	Critical scenarios pass
Document processing	Fixture-based and malformed-file tests	Supported files process; bad files fail safely
Extraction/entity resolution	Labelled fixtures and evaluation scripts	Metrics and error categories documented
Claim verification	Per-class metrics and adversarial cases	Confusion matrix and error analysis reported
Risk/graph	Baselines and planted-pattern tests	Data limitations clearly stated
Frontend	Manual and optional Playwright E2E	Core user journeys complete
Security	Authorization, upload validation and secret scan	No known critical demo issue left unresolved

13.3 Environments

Development: local Python environment and database. Test: synthetic/curated fixtures and isolated test database. Demo: Dockerized deployment if feasible. Production-grade operation requires further security, legal and operational review.

14. Security and Compliance

14.1 Threat model

Threat	Impact	Mitigation
Malicious uploaded file	Parser abuse or resource exhaustion	Type/size limits, timeouts, patched libraries, isolated processing where feasible
Unauthorized access	Supplier/business data exposure	Authentication, least privilege and authorization tests
Prompt injection in sources	Untrusted content affects model behaviour	Treat retrieved content as data; separate instructions and evidence; verify claims independently
Hallucinated explanation	Poor decision from unsupported claim	Evidence links, uncertainty labels, claim checks and human review
False entity match	Unrelated entities incorrectly linked	Conservative thresholds, explanation and manual review
Secret/data leakage	Credentials or private records exposed	No secrets in repo, minimal logs, protected storage and deletion process
Misleading feedback/data	Biased or unreliable model	Track provenance; version data; do not blindly train on feedback

14.2 AuthN/AuthZ

Implement secure authentication and API-level access checks. Roles may include user and reviewer/admin; access to documents and reports must be limited to authorized users.

14.3 Audit and logging

Record upload, analysis, report-access and review events while avoiding unnecessary raw document content or full sensitive identifiers.

14.4 Compliance and responsible use

This is an academic decision-support prototype. It is not a legal determination, guarantee of supplier reliability or substitute for human due diligence. Data must be used with permission and according to applicable terms.

15. Delivery and Operations

15.1 Release plan

Release	Scope	Exit criterion
Alpha	Profiles, intake, extraction and consistency checks	Upload-to-profile flow demonstrated
Beta	Evidence retrieval, claim verification, baseline risk indicators and report	Evidence-linked report and initial evaluation
Research release	Graph experiment, comparisons and error analysis	Reproducible evaluation documented
Final demo	Integrated UI/API, security checks, docs and presentation	Core acceptance criteria pass; limitations disclosed

15.2 CI/CD and rollback

Use GitHub for version control and optional GitHub Actions for linting/tests. Pin dependencies, document Docker builds, separate configuration from code and document rollback to a known-good version.

15.3 Monitoring and alerting

- Track processing failures, API errors, latency and job status.
- Record model/rule version and evidence timestamps for each assessment.
- Use basic logs for the academic demo; do not claim production monitoring until implemented.

15.4 Runbooks

Processing failure: Inspect error class, validate safely, and retry only when appropriate.

Poor extraction: Inspect OCR/source page and flag for manual correction.

Unsupported claim: Show insufficient evidence and request more sources or human review.

Unexpected risk spike: Inspect indicators, source freshness, model version and possible false matches.

15.5 Communication plan

Maintain a weekly progress log, mentor reviews, an updated GitHub backlog and milestone demonstrations.

16. Risks and Mitigations

16.1 Risk register

Risk	Likelihood/impact	Mitigation
No labelled fraud dataset	High/High	Data audit early; use clearly labelled synthetic benchmark; avoid unsupported predictive claims.
Scope too large for solo work	High/High	Build phases; defer GNN, integrations and monitoring if core flow slips.
OCR errors	High/Medium	Test varied quality; retain provenance; allow human review.
False entity match	Medium/High	Conservative thresholds and evidence-backed review; never label fraud from a shared attribute alone.
Verifier misses errors	Medium/High	Labelled evaluation, adversarial cases, uncertainty state and limitations.
Sources unavailable/restricted	Medium/Medium	Use permitted sources and user-provided docs; do not imply live verification.
Compute/deployment constraints	Medium/Medium	Local-first development, compact models and reproducible local demo.
Privacy/secret leakage	Low–Medium/High	Synthetic demo data, access controls, secret scanning and deletion process.

17. Research and Evaluation

17.1 Research questions

1. How accurately can selected supplier fields be extracted from heterogeneous documents, and which fields are most error-prone?
2. Do cross-document consistency checks improve detection of known discrepancies over isolated extraction?
3. How well does claim-level verification distinguish supported, contradicted and insufficient-evidence claims?
4. Can supplier graphs surface planted or labelled suspicious patterns with understandable false-positive rates?
5. How does evidence-grounded generation compare with an unverified LLM answer in factual support and traceability?

17.2 Experimental design

- Document extraction: label permitted/synthetic documents; report per-field precision, recall, F1 and error categories.
- Consistency: create matching and controlled-conflict pairs; compare rule-only and normalized entity-resolution pipelines.
- Claim verification: label claims against evidence; compare baseline LLM/RAG with retrieval plus independent verification.
- Risk scoring: use supervised models only if labels/sample size are defensible; otherwise evaluate transparent rules without claiming predictive fraud accuracy.

- Graph analysis: use synthetic graphs with planted relationships if real labels are unavailable; report this limitation.
- Ablation: disable individual components to estimate their contribution to selected metrics.

17.3 Baselines and metrics

Potential baselines include rule/regex extraction, simple retrieval, unverified LLM summaries and standard tabular classifiers when suitable labels exist. Metrics include precision, recall, F1, confusion matrices, PR-AUC/ROC-AUC where appropriate, evidence coverage, false-positive rate, latency and reviewer findings.

17.4 Expected outcomes

- Working pipeline for document intake, profiles, extraction, consistency checks, evidence retrieval, claim verification and traceable reports.
- Documented risk-indicator framework and graph-analysis experiment.
- Reproducible evaluation with baselines, failure analysis and dataset limitations.
- GitHub repository with setup instructions, tests, synthetic examples, architecture and demo.

17.5 Limitations

Synthetic data cannot establish real-world fraud performance. Public data may be stale or misattributed; OCR/entity resolution can fail; NLI models can misjudge nuanced claims. A risk score is not a probability of fraud unless calibrated and validated on representative data. These limits will be reported.

17.6 Future scope

- Additional permitted registry/certification connectors.
- Monitoring for source changes and certificate expiry.
- Audited reviewer feedback and supplier response workflows.
- More languages and document types.
- ERP/procurement integrations and multi-tenant SaaS hardening after security and market validation.

18. Appendices

18.1 Glossary

Term	Meaning
OCR	Optical Character Recognition.
NLP	Natural Language Processing.
RAG	Retrieval-Augmented Generation.
NLI	Natural Language Inference.
Entity resolution	Determining whether records refer to the same real-world entity.
GNN	Graph Neural Network.
PR-AUC	Area under the precision–recall curve.
Evidence provenance	Origin, time and context of evidence used in an assessment.

Technology stack: Python, FastAPI, React, TypeScript, Tailwind CSS, PostgreSQL, PyMuPDF/pdfplumber, PaddleOCR or Tesseract, spaCy, Hugging Face Transformers, sentence-transformers, FAISS, Scikit-learn, optional XGBoost/LightGBM, NetworkX, optional PyTorch Geometric, pytest, Docker, Git and GitHub.

References: Starting points for the literature review; verify exact editions and add recent peer-reviewed work on supplier-risk prediction, document fraud, graph fraud detection, hallucination evaluation and NLI claim verification.

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